JAVA Talk 4. Prym variety case

$\dim_{\mathbb{C}} A = n = 2m$

n	2	4	6	8	10	$2m$
$\dim_{\mathbb{C}} A_{HW}$	1	4	9	16	25	m^2
3:1 unramified cover	C_4/C_2	C_7/C_3	C_{10}/C_4	C_{13}/C_5	C_{16}/C_6	C_{3m+1}/C_{m+1}
4:1 unramified cover	$C_5/C_3/C_2$	$C_9/C_5/C_3$	$C_{13}/C_7/C_4$	$C_{17}/C_9/C_5$	$C_{21}/C_{11}/C_6$	$C_{4m+1}/C_{2m+1}/C_{m+1}$
$\dim \mathcal{P}_n^{\mathbb{Q}(\zeta_3) \text{ or } \mathbb{Q}(i)}$	3	6	9	12	15	$3m$

Rmk. By the above table (and a calculation of $H_{K\text{-basis}}$ *),
the Prym example only solves for
$n \leq 6$, $K = \mathbb{Q}(i)$ or $\mathbb{Q}(\zeta_3)$, $\det H_{K\text{-basis}} = (-1)^m$.

1. some Prym varieties are of Weil type
2. construction of alg cycles.

1. some Prym varieties are of Weil type

Setting

$\pi: \tilde{C} \longrightarrow C$ 3:1 unramified cover, $K = \mathbb{Q}(\zeta_3) = \mathbb{Q}(\sqrt{-3})$.

$\tilde{J} = \mathrm{Jac}(\tilde{C})$, $J = \mathrm{Jac}(C)$, $A = \mathrm{Prym}(\tilde{C}/C)$

(co)tangent space

$$T_0^*\tilde{J} \cong \underbrace{T_0^*J}_{\text{trivial part}} \oplus \underbrace{T_0^*A}_{\text{primitive part}}$$

i.e.,

$$H^0(\omega_{\tilde{C}}) \cong \underbrace{H^0(\omega_C)}_{\substack{\sigma\cdot s = s \\ \dim m+1}} \oplus \underbrace{H^0(\omega_C \otimes \eta)}_{\substack{\sigma\cdot s = \zeta_3 \cdot s \\ \dim m}} \oplus \underbrace{H^0(\omega_C \otimes \eta^2)}_{\substack{\sigma\cdot s = \zeta_3^2 \cdot s \\ \dim m}}$$

$\Rightarrow$ A is of Weil type.

η is an eigensheaf of $\mathbb{Z}/3\mathbb{Z}$-action, and
$\pi_*\mathcal{O}_{\tilde{C}} \cong \mathcal{O}_C \oplus \eta^{-1} \oplus \eta^{-2}$ is a dec into eigensheaves.

lattice:

$$H_1(\tilde{J};\mathbb{Q}) \cong \underbrace{H_1(J;\mathbb{Q})}_{\text{trivial part}} \oplus \underbrace{H_1(A;\mathbb{Q})}_{\text{primitive part}}$$

$n=2$ case:

$\tilde{\lambda}_1, \dots, \tilde{\lambda}_4$, $\quad \frac{1}{3}(\tilde{\lambda}_1 + \tilde{\lambda}_2 + \tilde{\lambda}_3)$, $\frac{1}{3}\tilde{\lambda}_4$

$\tilde{\lambda}_5, \dots, \tilde{\lambda}_8$ $\quad \frac{1}{3}(\tilde{\lambda}_5 + \tilde{\lambda}_6 + \tilde{\lambda}_7)$, $\tilde{\lambda}_8$

$\pi^*\omega = 3\tilde{\omega}$

$\lambda_1 := \tilde{\lambda}_1 - \tilde{\lambda}_2 \quad \lambda_3 := \tilde{\lambda}_1 + \tilde{\lambda}_2 - 2\tilde{\lambda}_3$

$\lambda_2 := \tilde{\lambda}_5 - \tilde{\lambda}_6 \quad \lambda_4 := \tilde{\lambda}_5 + \tilde{\lambda}_6 - 2\tilde{\lambda}_7$

$\{\lambda_1, \lambda_2\}$ is a K_{e_0}-basis for $H_1(A;\mathbb{Q})$, and $J_U = \begin{pmatrix} & & -3 & \\ 1 & & & -3 \\ & 1 & & \\ & & 1 & \end{pmatrix}$, $J_D = \begin{pmatrix} & 2 & & \\ -2 & & & \\ & & & 6 \\ & & -6 & \end{pmatrix}$,

$$H_\lambda = \begin{pmatrix} & 2\sqrt{-3} & & -6 \\ -2\sqrt{-3} & & 6 & \\ & 6 & & 6\sqrt{-3} \\ -6 & & -6\sqrt{-3} & \end{pmatrix}, \quad H_{K\text{-basis}} = \begin{pmatrix} & 2\sqrt{-3} \\ -2\sqrt{-3} & \end{pmatrix}, \quad \det H = 4\cdot(-3) \equiv -1$$

$\det H = -1$ is compatible with signature $(1,-1)$.
Some people use $(-1)^m \det H$ as signature.

2. construction of alg cycles

$$\begin{tikzcd}
\tilde{\tilde{S}}_1 \cup \tilde{\tilde{S}}_2 \cup \tilde{\tilde{S}}_3 \arrow[d] & \subset & C^{2m}_{3m+1} \arrow[d] & & \\
\tilde{S}_1 \cup \tilde{S}_2 \cup \tilde{S}_3 \arrow[d] & \subset & \mathrm{Sym}^{2m} C_{3m+1} \arrow[r, two heads, "AP^{2m}"] \arrow[d, "\text{ramified } 3^{2m-1}:1"] & A & \\
S_1 \cup S_2 \cup S_3 \arrow[d] & \subset & T \arrow[d, "\text{unramified } 3:1"] & & \\
\mathbb{P}^m \cong |\omega_{C_{m+1}}| & \subset & \mathrm{Sym}^{2m} C_{m+1} & &
\end{tikzcd}$$

$$\rightsquigarrow [AP^{2m}(\tilde{S}_i)] \in CH^m(A) \longrightarrow H^n(A;\mathbb{C}) \cap H^{m,m} = B^m(A)$$

Claim: For all Prym varieties A with specified e_0-action,

$$HW(A,\eta) \subseteq \langle [AP^{2m}(\tilde{S}_i)] \rangle_{\mathbb{Q}\text{-v.s, all } i}$$

No proof is presented here. For further reading:

$$\begin{tikzcd}
 & \tilde{C}^n \arrow[d, "\Xi"] \arrow[dr, "\mathcal{U}''"] & & \\
A \arrow[r, hook, "j"] & \tilde{J} \arrow[r, two heads, "p"] & A \arrow[r, "\mathcal{U}'"] & {}
\end{tikzcd}$$

$\mathcal{U}' := HW(A,\eta)$

$\mathcal{U}'' := p^*\mathcal{U}'$

$\mathcal{U} := \Xi^*\mathcal{U}''$

$\mathcal{U}'$ is alg

$\Leftrightarrow \mathcal{U}''$ is alg

$\Leftrightarrow \mathcal{U}$ is alg [Schoen 88, Thm 2.0]

Claim:

$$\mathcal{U} \subseteq \langle [\tilde{\tilde{S}}_i] \rangle_{\mathbb{Q}\text{-v.s. all } i}$$